NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

Multiparticle Quantum Mechanics: a Mathematical Tour to Second Quantization

Su Youyou (U2420256E)

Abstract

When we first learn quantum mechanics, we study frameworks such as the Schrödinger equation, Heisenberg equation, and path integral formulation. In these cases, we often describe systems by assigning labels to individual particles; however, in practice, we cannot always distinguish each particle. For example, when we have two electrons, there is no physical meaning in specifying which electron is at which position. Therefore, describing the system by assigning individual labels to particles is not a valid approach. In this project, we explore how second quantization provides another way to describe systems with identical particles, and investigate the relation between quantum field theory and stochastic processes through the Onsager–Machlup approach.

Hilbert Space for Multiparticle

For N distinguishable particles, the Hilbert space is the tensor product

$$\mathcal{H}^{\otimes N} = \mathcal{H} \otimes \mathcal{H} \otimes \cdots \otimes \mathcal{H}$$

The basis can be written as

$$|i_1, i_2, \ldots, i_N\rangle = |i_1\rangle \otimes |i_2\rangle \otimes \cdots \otimes |i_N\rangle$$

Note that for indistinguishable particles, we cannot use this.

Permutation on Tensor Product

Let π be any permutation on a group of N elements. For any

$$\pi \in S_N,$$

define

$$U_\pi : \quad \mathcal{H}^{\otimes N} \to \mathcal{H}^{\otimes N}$$

as

$$U_\pi(\psi_1 \otimes \cdots \otimes \psi_N) = \psi_{\pi^{-1}(1)} \otimes \cdots \otimes \psi_{\pi^{-1}(N)}$$

Bosonic Hilbert Space for N Particles

It is defined such that swapping any particles gives the same space.

$$\mathcal{H}_+^{(N)} = \{\Psi \in \mathcal{H}^{\otimes N} : U_\pi \Psi = \Psi \text{ for all } \pi \in S_N\}$$

Fermionic Hilbert Space for N Particles

It is defined such that swapping any even number of particles gives the same space while swapping odd number of particles gives the opposite of the space.

$$\mathcal{H}_-^{(N)} = \{\Psi \in \mathcal{H}^{\otimes N} : U_\pi \Psi = \operatorname{sgn}(\pi)\Psi \text{ for all } \pi \in S_N\}$$

Fock Space

Bosonic Fock Space is defined as

$$\mathcal{F}_+(\mathcal{H}) = \bigoplus_{N=0}^{\infty} \mathcal{H}_+^{(N)}.$$

Fermionic is defined as

$$\mathcal{F}_-(\mathcal{H}) = \bigoplus_{N=0}^{\infty} \mathcal{H}_-^{(N)}$$

Remark

Here, we don't know how many particles exactly we have, so we cannot decide which Hilbert space to use. Therefore, we direct sum up all Hilbert space and describe them in the Fock space. Note that the direct sum is actually a sequence of all Hilbert space of different number of particles.

Two-Particle Fock Space

For two bosons, swapping them gives the same state. So the wave function must satisfy

$$\Psi(x_1, x_2) = \Psi(x_2, x_1)$$

All vectors satisfying this compose a subspace, denoted as

$$\mathcal{H}_+^{(2)} = \operatorname{Sym}^2 \mathcal{H}.$$

For two fermions, the sign changes after swapping two of them

$$\Psi(x_2, x_1) = -\Psi(x_1, x_2)$$

All vectors satisfying this compose a subspace, denoted as

$$\mathcal{H}_-^{(2)} = \bigwedge^2 \mathcal{H}$$

Here, $\operatorname{Sym}^2 \mathcal{H}$ and $\bigwedge^2 \mathcal{H}$ are the symmetric and antisymmetric subspaces of $\mathcal{H} \otimes \mathcal{H}$.

Occupation Number States

For indistinguishable particles, we describe a state by the number of particles occupying each one-particle mode instead of labeling individual particles.
Choosing a basis $\{e_j\}$ of $\mathcal{H}$, the occupation state is

$$|n_1, n_2, \ldots\rangle,$$

where n_j denotes the number of particles in mode e_j.
For bosons:

$$n_j = 0, 1, 2, \ldots$$

For fermions:

$$n_j = 0, 1.$$

The total particle number is

$$N = \sum_j n_j.$$

Creation Operator

For a chosen mode e_j, the creation operator is

$$a_j^\dagger : \ \mathcal{F}_{+,\text{fin}}(\mathcal{H}) \to \mathcal{F}_{+,\text{fin}}(\mathcal{H})$$

where

$$a_j^\dagger |n_1, \ldots, n_j, \ldots\rangle = \sqrt{n_j + 1}\, |n_1, \ldots, n_j + 1, \ldots\rangle .$$

It adds the occupation number on e_j by 1.
Here $\mathcal{F}_{+,\text{fin}}$ is the Fock space of finite particles.

Annihilation Operator

For a chosen mode e_j, the annihilation operator is

$$a_j : \ \mathcal{F}_{+,\text{fin}}(\mathcal{H}) \to \mathcal{F}_{+,\text{fin}}(\mathcal{H})$$

where

$$a_j |n_1, \ldots, n_j, \ldots\rangle = \sqrt{n_j}\, |n_1, \ldots, n_j - 1, \ldots\rangle .$$

It subtracts the occupation number on e_j by 1.
Specifically,

$$a_j \Omega = 0$$

Field Operator

Recall that

$$|x\rangle = \sum_i |i\rangle \langle i|x\rangle = \sum_i \varphi_i^*(x) |i\rangle$$

where

$$\varphi_i(x) = \langle x|i\rangle$$

describes the amplitude of finding the particle at position x while the wave function is in state $|i\rangle$.
The field creation operator that creates a particle at position x is defined as

$$\hat{\psi}^\dagger(x) = \sum_i \varphi_i^*(x) a_i^\dagger.$$

Similarly, the field annihilation operator that annihilates a particle at position x is defined as

$$\hat{\psi}(x) = \sum_i \varphi_i(x) a_i.$$

Second Quantized Hamiltonian

One-body term:

$$\hat{H}^{(1)} = \int d^d x\, \hat{\psi}^\dagger \left(-\frac{\hbar^2}{2m} \nabla^2 + V \right) \hat{\psi}.$$

Interaction:

$$\hat{H}^{(2)} = \frac{1}{2} \int d^d x\, d^d y\, U(x - y) \hat{\psi}^\dagger(x) \hat{\psi}^\dagger(y) \hat{\psi}(y) \hat{\psi}(x).$$

Particle Number Conservation

$$\hat{N} = \int d^d x\, \hat{\psi}^\dagger(x) \hat{\psi}(x)$$

and

$$[\hat{N}, \hat{H}] = 0.$$

Therefore Hamiltonian evolution preserves particle number.

Multiparticle Heisenberg Equation

The Heisenberg equation for field annihilation operator

$$i\hbar \frac{\mathrm{d}\hat{\psi}(x,t)}{\mathrm{d}t} = \left(-\frac{\hbar^2}{2m} \nabla_x^2 + \hat{V}(x) \right) \hat{\psi}(x,t) + \int dy\, U(x - y) \hat{\psi}^\dagger(y,t) \hat{\psi}(y,t) \hat{\psi}(x,t).$$

Quantum Field Path Integral

Using coherent states,

$$\hat{a}_i |\alpha\rangle = \alpha_i |\alpha\rangle ,$$

the second-quantized Hamiltonian becomes a functional of complex fields.
The propagator takes the form

$$K = \int \mathcal{D}\varphi^* \mathcal{D}\varphi\, e^{\frac{i}{\hbar} S[\varphi^*, \varphi]}.$$

The integration variable changes from particle trajectories $x(t)$ to field configurations $\varphi(x,t)$, allowing the path integral to include all particle-number sectors in Fock space.

Onsager–Machlup Approach

The Langevin equation

$$\gamma \dot{x} = F(x) + \eta(t)$$

describes stochastic dynamics with Gaussian noise. The noise distribution induces a probability measure over trajectories:

$$P[x] \propto e^{-S_{\text{OM}}[x]},$$

where

$$S_{\text{OM}} = \frac{1}{4D} \int dt\, (\dot{x} - F(x))^2.$$

Thus, stochastic trajectories are described by an action functional, similar to the Feynman path integral.

Parisi–Wu Stochastic Quantization

The Euclidean quantum field path integral is

$$Z = \int \mathcal{D}\varphi\, e^{-S_E[\varphi]}.$$

Parisi and Wu introduced a fictitious stochastic time τ and proposed

$$\partial_\tau \varphi(x,\tau) = -\frac{\delta S_E}{\delta \varphi(x,\tau)} + \eta(x,\tau).$$

In the long-time limit,

$$P[\varphi] \propto e^{-S_E[\varphi]},$$

so the quantum field measure can be interpreted as the equilibrium distribution of a stochastic process.

Main Idea

Second quantization replaces the question

"Where is particle i?"

by

"How many particles occupy each mode?"

This framework naturally extends ordinary quantum mechanics to quantum field theory and reveals its connection with stochastic processes.
Full notes available at
`https://github.com/suyouyoualan-sketch/Odyssey-Second-Quantization/tree/main`

Supervisor: **Assoc. Prof. François Gay-Balmaz**